

**Sem 1 Examination 2015**  
**Question/answer booklet**

**YEAR 8**  
**MATHEMATICS**

**Name:** \_\_\_\_\_

**Teacher:** \_\_\_\_\_

**Section One:**  
**Calculator-free**

**Time allowed for this section**

**Section One**

Working time for paper: 20 minutes

**Material required/recommended for this paper**

**To be provided by the supervisor**

This question/answer Booklet

Formula sheet

**To be provided by the student**

*Standard items:* pens, pencils, pencil sharpener, eraser, correction fluid/tape, ruler, highlighter

*Standard items:* nil

**Important note to students**

This test comprises of **TWO** sections. The **first section** is **calculator free** where no calculators of any kind are to be used. The **second section** is **calculator assumed** where a scientific calculator may be used. All questions must be answered in both sections. **Answers should be rounded appropriately.** All working should be shown in the space provided. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks.

No pen, pencils, highlights etc. may be used during reading time. This time is to be used to read through the assessment and check that you understand what is being asked of you. You may speak with the teacher/supervisor during this time (by putting up your hand and waiting patiently for them to approach you) but you may only ask clarification questions and not how to solve the problems. After reading time has ended, you may not ask any more questions.

Multiple Choice shade in the correct answer

1. (11 Marks)

a. What is the answer to  $5 \times 7$ 

|                       |                       |                       |                                  |                       |
|-----------------------|-----------------------|-----------------------|----------------------------------|-----------------------|
| 2                     | 12                    | 7                     | 35                               | 25                    |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> |

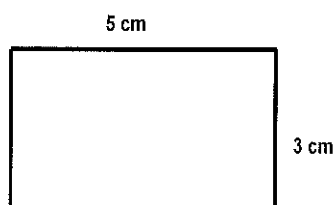
b. What is the value of  $35 - 7$  ?

|                       |                                  |                       |                       |                       |
|-----------------------|----------------------------------|-----------------------|-----------------------|-----------------------|
| 42                    | 28                               | 32                    | 17                    | 11                    |
| <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

c. How many metres are there in a kilometre?

|                       |                       |                                  |                       |                       |
|-----------------------|-----------------------|----------------------------------|-----------------------|-----------------------|
| 10                    | 100                   | 1000                             | 10000                 | 100 000               |
| <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Questions d and e relate to the diagram below



d. The perimeter of this rectangle above

|                       |                       |                       |                       |                                  |
|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------------|
| 15 cm                 | 8cm                   | 30 cm                 | 12 cm                 | 16cm                             |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> |

e. The area of the rectangle above is

|                                  |                       |                       |                       |                       |
|----------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| $15 \text{ cm}^2$                | $16 \text{ cm}^2$     | $20 \text{ cm}^2$     | $8 \text{ cm}^2$      | $35 \text{ cm}^2$     |
| <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

f. If the length of a parallelogram's base was 10cm and its height was 4 cm what would its area be ?

|                       |                       |                       |                                  |                       |
|-----------------------|-----------------------|-----------------------|----------------------------------|-----------------------|
| 40cm                  | 14m                   | 28cm                  | 40 cm <sup>2</sup>               | 28cm <sup>2</sup>     |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> |

g.  $-5 + 7 =$

|                                  |                       |                       |                       |                       |
|----------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 2                                | -12                   | -35                   | -2                    | 12                    |
| <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

h  $4 + 2 \times 3 =$

|                       |                                  |                       |                       |                       |
|-----------------------|----------------------------------|-----------------------|-----------------------|-----------------------|
| 63                    | 10                               | 18                    | 24                    | 12                    |
| <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

i What is the formula used to calculate the area of a circle

|                       |                       |                       |                       |                                  |
|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------------|
| $A = \pi \cdot D$     | $A = 2\pi r$          | $A = \pi \cdot L$     | $A = \pi \cdot D^2$   | $A = \pi \cdot r^2$              |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> |

j  $3^2 \cdot 3^4$

|                                  |                       |                       |                       |                       |
|----------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| $3^6$                            | $3^8$                 | $9^8$                 | $6^6$                 | 0                     |
| <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

k If  $h = 4a - 3b$  and  $a = 5$  and  $b = 2$  what is the value of h

|                       |                       |                       |                       |                                  |
|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------------|
| 13                    | 12                    | 26                    | 15                    | 14                               |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> |

2. (10 marks)

Work out the following and write your answer in the box on the right

a.  $9 - 5 =$

Answer  
here

4

b.  $-3 - 4 =$

-7

c.  $-8 + 5 =$

-3

d.  $12 + (-15) =$

-3

e.  $4 \times 4 =$

16

f.  $-5 \times 2 =$

-10

g.  $-1 \times -3 =$

3

h.  $20 \div 5 =$

4

i.  $12 \div -4 =$

-3

j.  $-8 \div 2 =$

-4

3. (2 marks: 1,1)

Use the index laws to find the following as simple powers

a.  $6^5 \times 6^3$

$6^8$

b.  $4^5 \div 4^3$

$4^2$

4. (5 marks 1,1,1,2)

Simplify

a.  $2a + 3a$

$5a$

b.  $7b - 5b$

$2b$

c.  $2c + 4j + 3c - 2j$

$5c + 2j$

d.  $3xy + 4x + 5yx + 6y - 2xy$

$6xy + 4x + 6y$

5. Expand the following (4 marks: 2,2)

a.  $2(x + 3)$

$2x + 6$

b.  $c(a + b)$

$ac + bc$